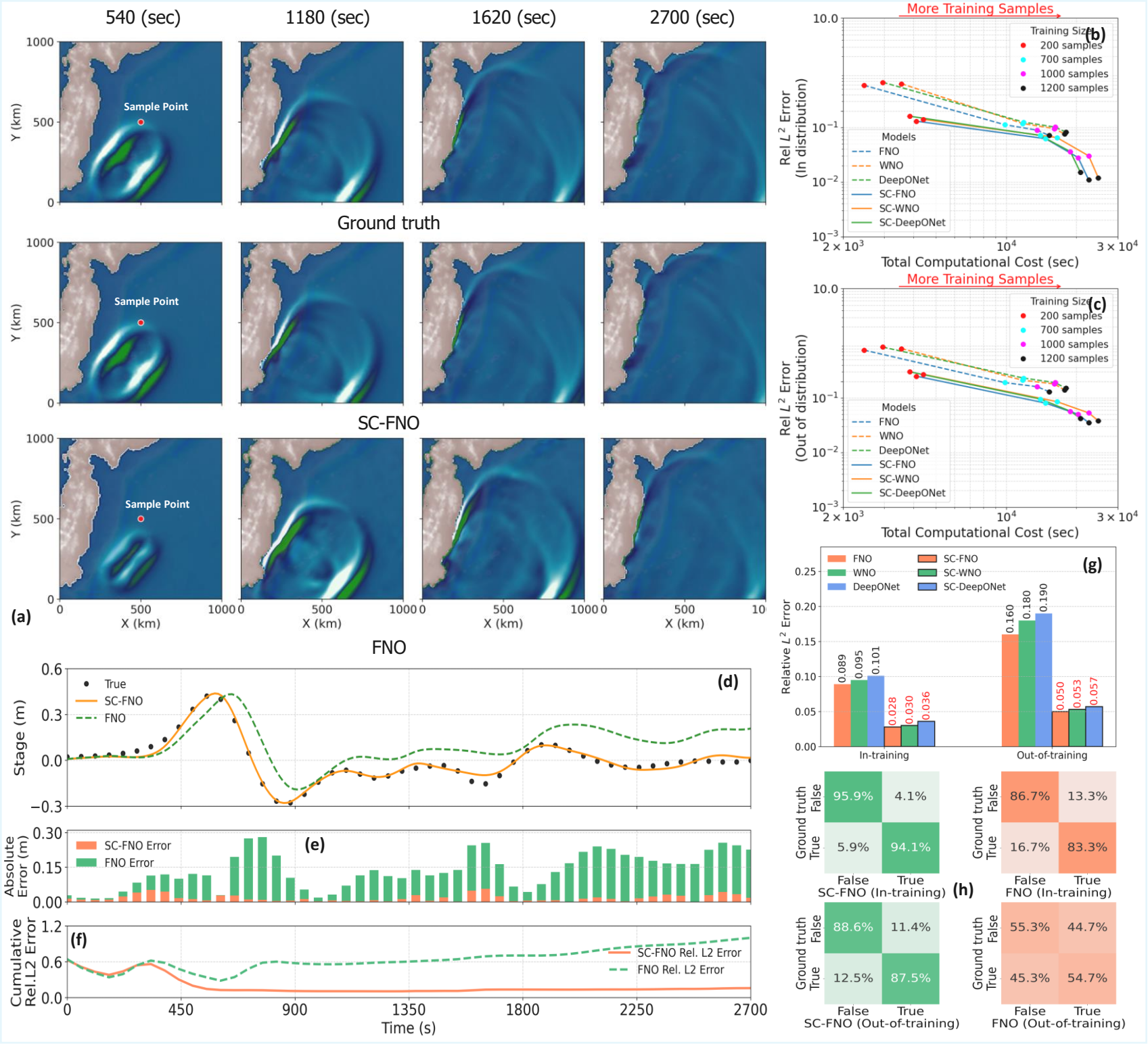
540 (sec)
1180 (sec)
1620 (sec)
2700 (sec)
Sample Point
Ground truth
SC-FNO
FNO
Y (km)
X (km)
(a)
More Training Samples
Training Size
200 samples
700 samples
1000 samples
1200 samples
Models
FNO
WNO
DeepONet
SC-FNO
SC-WNO
SC-DeepONet
Rel L^2 Error (In distribution)
Rel L^2 Error (Out of distribution)
Total Computational Cost (sec)
(b)
(c)
(g)
Relative L^2 Error
In-training
Out-of-training
0.089
0.095
0.101
0.028
0.030
0.036
0.160
0.180
0.190
0.050
0.053
0.057
True
SC-FNO
FNO
(d)
Stage (m)
(e)
SC-FNO Error
FNO Error
Absolute Error (m)
(f)
Cumulative Rel.L2 Error
SC-FNO Rel. L2 Error
FNO Rel. L2 Error
Time (s)
Ground truth
False
True
95.9%
4.1%
5.9%
94.1%
SC-FNO (In-training)
86.7%
13.3%
16.7%
83.3%
FNO (In-training)
(h)
88.6%
11.4%
12.5%
87.5%
SC-FNO (Out-of-training)
55.3%
44.7%
45.3%
54.7%
FNO (Out-of-training)